

$$\operatorname{Re}(\langle u, v \rangle) = \frac{1}{4} (\|u+v\|^2 - \|u-v\|^2)$$

$$\operatorname{Im}(\langle u, v \rangle) = \frac{1}{4} (\|u-iv\|^2 - \|u+iv\|^2)$$

$$\|u+v\|^2 = \langle u+v, u+v \rangle = \langle u+v, u \rangle + \langle u+v, v \rangle = \langle u, u \rangle + \langle u, v \rangle + \langle v, u \rangle + \langle v, v \rangle =$$

$$= \|u\|^2 + \|v\|^2 + \langle u, v \rangle + \overline{\langle u, v \rangle}$$

$$\|u-v\|^2 = \|u\|^2 + \|v\|^2 + \langle u, -v \rangle + \overline{\langle u, -v \rangle}$$

$$\|u+v\|^2 - \|u-v\|^2 = \langle u, v \rangle + \overline{\langle u, v \rangle} - \langle u, -v \rangle - \overline{\langle u, -v \rangle} =$$

$$= 2\langle u, v \rangle + 2\overline{\langle u, v \rangle}$$

Since $2\operatorname{Re}(\langle u, v \rangle) = x + \bar{x}$

$$4\operatorname{Re}(x) = 2x + 2\bar{x}$$

$$4\operatorname{Re}(\langle u, v \rangle) = 2\langle u, v \rangle + 2\overline{\langle u, v \rangle} = \|u+v\|^2 - \|u-v\|^2$$

$$\operatorname{Re}(\langle u, v \rangle) = \frac{1}{4} (\|u+v\|^2 - \|u-v\|^2)$$

$$\|u-iv\|^2 = \|u\|^2 + \|v\|^2 + \langle u, -iv \rangle + \langle -iv, u \rangle = \|u\|^2 + \|v\|^2 + i\langle \overline{u}, v \rangle - i\langle u, \overline{v} \rangle$$

$$\|u+iv\|^2 = \|u\|^2 + \|v\|^2 + i\langle u, v \rangle - i\langle \overline{u}, \overline{v} \rangle$$

$$\|u-iv\|^2 - \|u+iv\|^2 = -2i(\langle u, v \rangle - \langle \overline{u}, \overline{v} \rangle)$$

$$4i \operatorname{Im}(x) = 2x - 2\overline{x}$$

$$4i \operatorname{Im}(\langle u, v \rangle) = 2(\langle u, v \rangle - \langle \overline{u}, \overline{v} \rangle) = \frac{1}{-i} (\|u-iv\|^2 - \|u+iv\|^2)$$

$$4i \operatorname{Im}(\langle u, v \rangle) = \frac{1}{-i} (\|u-iv\|^2 - \|u+iv\|^2)$$

$$\operatorname{Im}(\langle u, v \rangle) = \frac{1}{4} (\|u-iv\|^2 - \|u+iv\|^2)$$